Full-Pitch Tracking of a Thrust-Vectoring Aerial Manipulator: A Simulation Study of Equivalent Coupling Effects

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***Abstract*—Near-wall manipulation motivates aerial platforms that accommodate arm reach without bringing the airframe unnecessarily close to a surface. This paper studies a thrust-vectoring aerial manipulator with four tiltable propulsion units and a five-degree-of-freedom arm, focusing on full-pitch tracking under prescribed mass-property variations. Thrust reorientation provides a design basis for vertical-surface operations, although workspace and aerodynamic benefits are not evaluated here. Near a 90° pitch attitude, differential-thrust pitch authority diminishes, while nonlinear thrust–moment allocation and changing mass distribution complicate tracking. The control architecture combines quaternion attitude feedback, filtered proportional–integral–derivative rate control, and fixed-gain extended state observers for equivalent-disturbance estimation and compensation. A nonlinear allocator coordinates rotor thrusts and differential tilt to target three moments and a collective-lift projection. Conditional continuous-time analysis establishes a finite-time bound for an auxiliary local pitch model under uniform residual bounds, a suitable Lyapunov family, and an invariant-domain condition; its correspondence with the two-branch digital controller is not established. Paired simulations evaluate a 0.2 rad/s pitch revolution with a 3-cm forward and 2-cm body-down center-of-mass shift and a 10% inertia increase. Under complete-reference replay and online reference generation, perturbed peak geodesic errors of the simulated attitude estimates over 20–65 s are 4.833° and 4.886°, respectively, while position-error peaks are 2.406 m and 0.2913 m. Both perturbed cases maintain peak attitude-estimate tracking errors below 5°, and their recorded attitude estimates subsequently traverse the complete revolution. With online reference generation, the position-error peak increases by only 0.0080 m relative to the nominal case; replay instead exhibits substantial lateral drift. These results provide encouraging preliminary numerical evidence of attitude-estimate tracking under the prescribed mass-property variations and highlight the role of reference feedback in position regulation. Completion delay and configuration-dependent position drift remain limitations.**

***Index Terms*—Aerial manipulation, thrust vectoring, quaternion attitude tracking, control allocation, numerical evaluation.**

I. Introduction

VERTICAL-surface manipulation requires an aerial robot to position its end effector while maintaining a suitable airframe attitude and clearance. Arm reach and base motion are therefore coupled task requirements [1], [2]. For conventional fixed-axis multirotors, the coupling between thrust direction and attitude constrains the base configurations available for manipulation [3]. A limited arm workspace can consequently require closer airframe placement. This motivates platforms that can reorient the airframe while retaining the force authority needed for position regulation.

Tiltable propulsion provides this capability by changing thrust directions relative to the airframe. Overactuated and omniorientational vehicles demonstrate the value of this additional actuation freedom [4]–[6]. More directly relevant to vertical configurations, a quad tilt-rotor has demonstrated a seamless 90° attitude transition under position control [7]. These results establish a technical basis for selecting manipulation-friendly attitudes, but the attainable wrench remains dependent on rotor geometry and tilt configuration. The present platform has four propulsion units tilting about axes parallel to the body pitch axis and a five-degree-of-freedom arm; its direct thrust is confined to the body x–z plane.

For this coplanar geometry, thrust reorientation during a pitch maneuver changes the relative effectiveness of rotor-speed and tilt commands. As the thrust direction approaches the body longitudinal direction near vertical attitude, the pitch moment generated by differential rotor thrust diminishes, increasing the importance of coordinated tilt allocation. Meanwhile, changes in arm configuration alter the center of mass and inertia, modifying the moments and accelerations that the controller must accommodate. The problem is thus to maintain tracking as both actuation effectiveness and equivalent coupling vary. Here, this problem is examined over a complete pitch revolution through the vertical configurations, using prescribed mass-property variations rather than an identified joint-motion model. Neither wall aerodynamics nor contact interaction is included.

Existing methods address complementary parts of this problem. Geometric attitude control avoids a single Euler-angle chart for large rotations [8], [9], whereas constrained control allocation addresses the conversion of requested forces and moments into actuator commands [10]. For aerial manipulation, Cao et al. combine extended state observers (ESOs), flight control, and cooperative planning to estimate and compensate coupling on a quadcopter–Delta-arm platform [11]. Disturbance estimation reduces dependence on a detailed coupling model, but cannot itself restore unavailable actuator authority. Applying such a strategy to the present geometry therefore also requires configuration-dependent allocation and an explicit treatment of allocation mismatch in the analysis.

The present architecture combines quaternion attitude feedback, filtered proportional–integral–derivative rate control, fixed-gain ESOs, and nonlinear allocation. Two fixed-gain rate PID branches are mixed using a thrust-direction-dependent weight after branch saturation; ESO compensation enters the second branch. The allocator coordinates four thrusts and

one differential tilt to target three moments and a collective-lift projection, while the ESOs estimate equivalent rate disturbances relative to their internal branch inputs. Complete-reference replay and online reference generation are evaluated separately because reference feedback also affects position regulation. A complete pitch task additionally requires path evidence: its initial and final physical attitudes coincide. Unwrapped pitch-related progress and ordered crossings therefore assess traversal separately from reference-end accuracy.

The contributions are threefold:

1) A simulation formulation for the planar-tilt aerial manipulator combines geometry-consistent allocation with prescribed center-of-mass and inertia variations, distinguishing moving-center corrections from the equivalent input applied to a fixed-inertia plant.
2) A conditional finite-time bound is derived for an idealized continuous-time local pitch model, retaining filtered rate-error and integral states. The result requires uniform residual bounds, a suitable scheduling-dependent Lyapunov family, and a self-consistent invariant domain; it does not certify full-system or digital-implementation stability.
3) Paired simulations quantify attitude-estimate tracking error, plant-position error, and estimated revolution progress under complete-reference replay and online reference generation. Peak geodesic attitude-estimate errors remain below 5° in both perturbed cases, and the online configuration shows a 0.0080-m increase in position-error peak relative to its nominal case. The recorded attitude estimates traverse the complete revolution after the reference ends. The replay configuration exposes substantial lateral drift, providing a contrasting case for interpreting the role of reference feedback.

Section II defines the platform, task, and equivalent model. Section III describes the control and allocation architecture. Section IV presents the conditional analysis, and Section V reports the paired simulations, followed by discussion and conclusions.

II. Platform, Task, and Equivalent Coupling Model

A. Frames and Propulsion Geometry

The motivating platform has an H-shaped airframe, four independently tiltable propulsion units, and a five-degree-of-freedom arm mounted on the airframe. The body frame uses forward–right–down axes; the inertial frame uses north–east–down axes. The scalar-first Hamilton quaternion q and rotation matrix $R = R(q)$ map body coordinates to inertial coordinates. With body angular velocity $\Omega = [\Omega_x, \Omega_y, \Omega_z]^T$,

$$\dot{R} = R\widehat{\Omega}, \qquad \dot{q} = \tfrac{1}{2} q \otimes [0, \Omega^T]^T. \tag{1}$$

Here $\widehat{\Omega} v = \Omega \times v$. Rotor speed is denoted by ω_i, separately from the body rates.

Let O be the fixed body/sensor origin whose position is logged and C the instantaneous equivalent center of mass. All unqualified force and moment components below are expressed in the body frame. A subscript O or C identifies the moment reference point. The simulation uses fixed thrust application points

$$\begin{aligned} r_1 &= (L, L, 0)^T, \quad & r_2 &= (-L, -L, 0)^T, \\ r_3 &= (L, -L, 0)^T, \quad & r_4 &= (-L, L, 0)^T, \end{aligned} \qquad L = \frac{0.318}{\sqrt{2}} \text{ m}. \tag{2}$$

Equation (2) defines the simulation rotor numbering used throughout this paper; implementation conventions are summarized in Appendix A.

For the physical planar-tilt geometry, a mathematical tilt δ_i about $+y_B$ gives the unit thrust direction

$$d_i = -R_y(\delta_i) e_3 = [-\sin\delta_i, 0, -\cos\delta_i]^T. \tag{3}$$

The quasi-static propulsion force and moment about O are

$$F_a = \sum_i T_i d_i, \qquad \tau_{a,O} = \sum_i \big(r_i \times T_i d_i - \chi_i c_\tau T_i d_i \big), \tag{4}$$

where $T_i = C_t \omega_i^2$, $c_\tau = C_m / C_t$, and χ_i follows the simulation spin convention. Since the thrust directions lie in the body x–z plane, $F_{a,y} = 0$, and the propulsion wrench can be represented by $[F_{a,x}, F_{a,z}, \tau_{a,O}^T]^T$. The control allocator targets three body moments and one collective-lift projection, as defined in Section III.

The specified mechanical tilt range is $\pm 150^\circ$. The numerical tilt-encoding limits are given separately in Table II.

B. Nominal Pitch Path and Complete Reference

The nominal path is a positive pitch revolution beginning at $t_r = 20$ s with $\omega_r = 0.2$ rad/s:

$$\theta_d(t) = \mathrm{clip}\big(\omega_r (t - t_r), 0, 2\pi\big), \tag{5}$$

$$q_{\mathrm{pitch}}(t) = q_0 \otimes [\cos(\theta_d/2), 0, \sin(\theta_d/2), 0]^T. \tag{6}$$

The analytical end is $t_r + 2\pi/\omega_r = 51.4159\ldots$ s; the sampled reference-end check uses $t_f = 51.416$ s. The pitch rate jumps at both clipping instants. This is a clipped constant-rate attitude command; the quintic profile introduced below prescribes mass-property variation only.

Equation (6) defines the nominal pitch path. The complete online reference $q_{d,\mathrm{on}}$ additionally includes the documented feedback-generated roll adjustment; $q_{d,\mathrm{rep}}$ denotes the recorded complete reference used for replay. The selected signal q_d enters the quaternion attitude loop in Section III. The complete reference is specified by (22); the nominal pitch path alone omits its roll component.

For the nominal path, $q_{\mathrm{pitch}}(t_f) = -q_0$ represents the same physical attitude as q_0. These endpoints alone do not determine the number of revolutions. Completion is assessed by accumulated y-twist progress and ordered phase crossings, with reference-end error evaluated separately in Section V.

C. Prescribed Equivalent Mass Properties

S1 uses constant mass properties. S2 enables a prescribed equivalent rigid-body perturbation, with no additional arm or payload mass. The total mass is $m = 2.288$ kg and

$$J_0 = \mathrm{diag}(0.082, 0.082, 0.149)\ \mathrm{kg\,m^2}. \tag{7}$$

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For $t \in [27, 45]$ s, let $u_m = (t-27)/T_{\rm arm}$ and $T_{\rm arm} = 18$ s. The profile is

$$s = 10u_m^3 - 15u_m^4 + 6u_m^5, \tag{8}$$
$$\dot{s} = 30u_m^2(1-u_m)^2/T_{\rm arm}, \tag{9}$$
$$\ddot{s} = 60u_m(1-u_m)(1-2u_m)/T_{\rm arm}^2. \tag{10}$$

Before 27 s, $(s, \dot{s}, \ddot{s}) = (0, 0, 0)$; after 45 s, it equals $(1, 0, 0)$. The center-of-mass offset and inertia about C are

$$c_B = [0.03, 0, 0.02]^T s \text{ m}, \tag{11}$$
$$J = (1 + 0.10s)J_0, \qquad \dot{J} = 0.10\dot{s}J_0. \tag{12}$$

The positive third offset is body down. The final offset and inertia increase persist after 45 s. Relative internal angular momentum is set to zero, $h_{\rm rel} = 0$. Thus, s is a mass-distribution proxy, not a joint trajectory or an identified multibody-arm model.

D. Moment Transfer and Fixed-Origin Dynamics

The positions satisfy $p_C^I = p_O^I + Rc_B$. Let $F_{\rm ng}$ be the non-gravitational resultant force and τ_O its associated actuator/environment moment about O. Gravity acts at C. Transfer to C gives

$$\tau_C = \tau_O - c_B \times F_{\rm ng}, \tag{13}$$
$$\tau_{\rm corr} = -c_B \times F_{\rm ng} - \dot{J}\Omega, \tag{14}$$
$$\alpha = J^{-1}\big[\tau_O + \tau_{\rm corr} - \Omega \times (J\Omega)\big]. \tag{15}$$

The combined correction $\tau_{\rm corr}$ contains the moment transfer and inertia-rate term exactly once.

Differentiating the position relation gives the fixed-origin correction

$$F_{\rm corr} = -m\big[\ddot{c}_B + 2\Omega \times \dot{c}_B + \alpha \times c_B + \Omega \times (\Omega \times c_B)\big], \tag{16}$$
$$m\ddot{p}_O^I = mg^I + R(F_{\rm ng} + F_{\rm corr}). \tag{17}$$

This correction arises from evaluating position at O; it is not an additional external wind force on the vehicle.

The retained plant integrates a fixed-J_0 rotational equation. The S2 module therefore injects

$$\tau_{\rm add} = J_0\alpha + \Omega \times (J_0\Omega) - \tau_O, \tag{18}$$

which makes $J_0\dot{\Omega} = \tau_O + \tau_{\rm add} - \Omega \times (J_0\Omega)$ reproduce $\dot{\Omega} = \alpha$. The quantities $\tau_{\rm corr}$ and $\tau_{\rm add}$ are in N m; $J_0^{-1}\tau_{\rm add}$ is the incremental angular acceleration in rad/s^2. The fixed-inertia rotational plant receives the total moment $\tau_O + \tau_{\rm add}$. The correction $\tau_{\rm corr}$ enters through α in (15) and is therefore not added separately. The translational plant receives $F_{\rm ng} + F_{\rm corr}$ to represent the motion of the fixed origin O.

III. Implemented Controller and Allocation

The controller uses simulated inertial-navigation estimates $\hat{p}_O$, $\hat{v}$, $\hat{q}$, and $\hat{\Omega}$. These are distinct from the plant states. Figure 1 summarizes the two-reference architecture. The controller, ESO, and plant update at $h = 0.002$ s; the position-command layer updates at $h_p = 0.004$ s. The simulation uses the fixed-step `ode1` solver with a 0.002-s step. Appendix A specifies discrete filters, initialization, and command encoding.

A. Position Feedback and Complete Reference

Let $C(\hat{\psi}) = R_z(-\hat{\psi})$ map inertial horizontal vectors to the heading frame. In position-hold mode, the horizontal target $p_{h,*}$ is latched from the estimated position on entry. The vertical target is $z_* = -\hat{h}(20)$ from the first active hold sample at 20 s. The velocity commands are

$$v_{h,c} = \mathrm{sat}_5\Big(0.95[C(\hat{\psi})(p_{h,*} - \hat{p}_h, 0)^T]_{1:2}\Big),$$
$$v_{z,c} = \mathrm{sat}_{2.5}(z_* - \hat{z}), \qquad \hat{z} = -\hat{h}. \tag{19}$$

The gains are in s^{-1} and velocity limits in m/s; saturation is componentwise. In all four cases, the hold targets are approximately $(0.013287, 0.048185)$ m horizontally and $\hat{h}(20) = 1.801811$ m vertically. These controller targets differ from the evaluation target in Section V, which uses $x_d = y_d = 0$ and the plant height $h(20) = 1.547105$ m.

Second-order command-tracking filters produce $\bar{v}_{h,c}$ and $\bar{v}_{z,c}$ from these commands (Appendix A-A). Define

$$e_{v,h} = \bar{v}_{h,c} - [C(\hat{\psi})\hat{v}]_{1:2}, \qquad e_{v,z} = \hat{v}_z - \bar{v}_{z,c}. \tag{20}$$

The horizontal velocity PID uses $(k_p, k_i, k_d) = (1.4, 0.2, 0.2)$, with integral and derivative contributions individually clipped to $[-1, 1]$. Its output $a_{h,c}$ has acceleration units. The vertical PID uses $(0.6, 0.1, 0)$, integral limits $[-0.15, 0.15]$, and output limits $[-0.35, 0.35]$; its output u_h is normalized. Both use the discrete PID realization in Appendix A-A, with a 10-Hz derivative filter. The horizontal integrator resets to zero; the vertical integrator resets to -0.15.

The horizontal acceleration command generates

$$\phi_c = \mathrm{sat}_{\pi/6}[\arctan(1.1a_{h,c,y}/9.8055)],$$
$$a_x = \mathrm{sat}_{\pi/6}[\arctan(-1.1a_{h,c,x}/9.8055)],$$
$$a_\ell = s_q a_x, \quad s_q = \begin{cases} -1, & \hat{q}_y > 1.414/2, \\ 1, & \text{otherwise.} \end{cases} \tag{21}$$

The additional pilot pitch-angle input is zero in the evaluated position-hold task. The sign test in (21) uses the signed quaternion component, not its absolute value. It is part of the implemented common-tilt logic.

With $q_x(\phi) = [\cos(\phi/2), \sin(\phi/2), 0, 0]^T$ and $q_y(\theta) = [\cos(\theta/2), 0, \sin(\theta/2), 0]^T$, the complete online reference is

$$q_{d,\rm on} = q_y(\theta_d) \otimes q_x(\phi_c). \tag{22}$$

The yaw command is zero. Replay supplies the complete online-S1 quaternion history to the same attitude input, bypassing the current lateral-error adjustment of ϕ_c. The common-tilt channel in (21) and vertical feedback remain active in both configurations. The collective PWM command is obtained by integer conversion of $1000 + 1000\,\mathrm{clip}(0.5 + u_h, 0, 1)$ and clipped to $[1100, 1900]$ at the allocator input. Thus P_0 remains a real-time altitude-loop output during replay.

B. Attitude and Parallel Rate Loops

Both attitude quaternions are normalized, with a 10^{-6} norm floor. The attitude error and its shortest-path representative are

$$q_e = \hat{q}^{-1} \otimes q_d = [\eta_e, \epsilon_e^T]^T, \qquad q_e^+ = \begin{cases} q_e, & \eta_e \geq 0, \\ -q_e, & \eta_e < 0. \end{cases} \tag{23}$$

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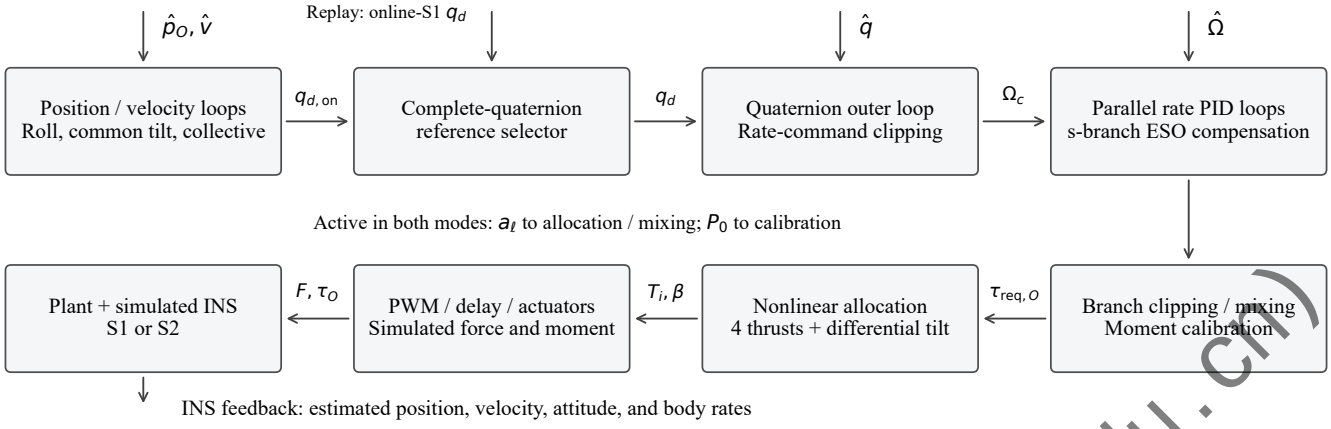


Fig. 1. Simulation control architecture. Position feedback generates the common-tilt command and online roll reference. Replay replaces only the complete desired quaternion with the online-S1 reference; common tilt and collective control remain active. Two parallel rate PID branches are combined after branch saturation, with ESO compensation applied to the second branch before saturation. All feedback is supplied by the simulated INS.

The implemented outer loop applies a gain of 7 to all three vector components and first clips to $\pm\pi$ rad/s. Rate-command preprocessing then clips every component to $\pm\pi/2$ rad/s. Equivalently,

$$\Omega_c = \mathrm{sat}_{\pi/2}\big(7\,\mathrm{sgn}(\eta_e^+)\epsilon_e^+\big)\,. \tag{24}$$

At $\eta_e^+ = 0$, the implemented sign block returns zero. There is no desired-rate feedforward; the separately configured yaw-rate input is not connected to the active rate-command output.

Let $e_\Omega = \Omega_c - \hat{\Omega}$. A shared 30-Hz low-pass filter supplies e_f to two parallel PID branches, denoted m and s:

$$\begin{aligned}
e_f[k+1] &= e_f[k] + h\omega_f(e_\Omega[k] - e_f[k]),\\
u_b[k] &= K_{p,b}e_f[k] + I_b[k] + D_b[k], \quad b \in \{m, s\},\\
I_b[k+1] &= \mathrm{sat}_{0.1}(I_b[k] + hK_{i,b}e_f[k])\,,\\
\xi_b[k+1] &= \xi_b[k] + h\omega_f(e_f[k] - \xi_b[k]),\\
D_b[k] &= \mathrm{sat}_{0.1}(K_{d,b}\omega_f(e_f[k] - \xi_b[k]))\,,
\end{aligned} \tag{25}$$

where $\omega_f = 2\pi(30)$ rad/s and each K is diagonal. The two integral states are separate. The derivative term uses an additional tracking filter; it is not simply $K_d\dot{e}_f$. PID outputs are normalized command quantities, not physical angular accelerations. Reset initializes $I_b = 0$ and $\xi_b = e_f$, and sets the derivative output to zero. Integral-output clipping provides the implemented windup protection; no allocator-residual back-calculation is used.

C. ESO Compensation and Branch Mixing

A fixed-gain ESO receives the compensated s-branch command before saturation:

$$\nu = u_s - \Gamma B_0^{-1} z_2, \qquad e_o = z_1 - \hat{\Omega}, \tag{26}$$

$$\begin{aligned}
z_1[k+1] &= z_1[k] + h(B_0\nu[k] + z_2[k] - B_1 e_o[k]),\\
z_2[k+1] &= \mathrm{sat}_{\bar{z}_2}(z_2[k] - hB_2 e_o[k])\,.
\end{aligned} \tag{27}$$

The diagonal gains and limits are listed in Table I. Reset sets $z_1 = \hat{\Omega}$ and $z_2 = 0$. The compensation divides only Γz_2 by B_0; it does not divide the PID output by B_0. The ESO input is not the final allocated command or the realized plant moment.

TABLE I
RATE-LOOP AND OBSERVER PARAMETERS (ROLL, PITCH, YAW)

Parameter	Value
$K_{p,m}$; $K_{i,m}$	$(0.1, 0.1, 0.15)$; $(0.1, 0.1, 0.2)$
$K_{d,m}$	$(0.003, 0.003, 0.001)$
$K_{p,s}$; $K_{i,s}$	$(0.1, 0.1, 0.15)$; $(0.04, 0.03, 0.15)$
$K_{d,s}$	$(0.002, 0.003, 0.001)$
B_0	$(300, 300, 200)$
B_1, B_2	$(500, 500, 350)$; $(1200, 1800, 1400)$
Γ	$(0.4, 0.5, 0.5)$
z_2 magnitude limits	$(60, 60, 100)$ rad/s^2
Rate-error / derivative cutoff	30 / 30 Hz
I_b, D_b limits	± 0.1 per component
Branch command limits	± 1 per component

Let $d_0 = Q_\ell(\hat{q})[-\sin a_\ell, 0, -\cos a_\ell]^T$. The command sent to allocation is

$$\begin{aligned}
\lambda_a &= \frac{d_{0,x}^2}{\max(d_{0,x}^2 + d_{0,z}^2, 10^{-12})},\\
u &= (1 - \lambda_a)\,\mathrm{sat}_1(u_m) + \lambda_a\,\mathrm{sat}_1(\nu).
\end{aligned} \tag{28}$$

This is an output mixture of two fixed-gain PID branches, with compensation in the second branch. The same weight acts on all three command axes. Together, (26), (28), and (32) specify the complete transformation from the rate controllers to the requested physical moment.

D. Moment Calibration and Signal Chain

The allocation interface takes a collective PWM command P_0, a normalized three-axis command u, a common lift-frame angle a_ℓ, and $\hat{q}$. Its physical moment target is computed as

$$\omega_0 = k_\omega(P_0 - 1000)10^{-3} + \omega_b, \qquad T_0 = C_t\omega_0^2, \tag{29}$$

$$g_T = 2C_t\omega_0 k_\omega 10^{-3}, \tag{30}$$

$$\gamma_\tau = 1000 g_T[4\sqrt{2}L, 4\sqrt{2}L, 4c_\tau]^T, \tag{31}$$

$$\tau_{\text{req},O} = \mathrm{diag}(\gamma_\tau)u. \tag{32}$$

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Here $P_0 \in [1100, 1900]$, g_T converts PWM increments to thrust increments, and γ_τ converts normalized commands to N m. The calibration depends on the collective command.

E. Five-Variable, Four-Output Allocation

The allocation decision and target are

$$z_a = [T_1, T_2, T_3, T_4, \beta]^T, \tag{33}$$

$$\mathcal{H}(z_a; a_\ell, \hat{q}) = \begin{bmatrix} \tau_{\mathrm{pred},O}(z_a; a_\ell, \hat{q}) \\ (\sum_i T_i)\cos\beta \end{bmatrix}, \tag{34}$$

$$y_* = [\tau_{\mathrm{req},O}^T, 4T_0]^T. \tag{35}$$

There are four residual components: three moments and one collective-lift projection. The common angle a_ℓ is supplied to the solver, whereas β is a decision variable representing opposite angular offsets for the two propulsion groups. The solver evaluates the differential-tilt contribution to the complete moment vector and lift projection. The two force entries of the geometric wrench in Section II are not independent targets.

The static predictor uses $\hat{R} = R(\hat{q})$ and a virtual gravity-aligned basis $Q_\ell(\hat{q})$:

$$d_0 = Q_\ell(\hat{q})[-\sin a_\ell, 0, -\cos a_\ell]^T, \tag{36}$$

$$d_i^{\mathrm{sim}} = R_y(\sigma_i \beta) d_0, \tag{37}$$

$$\tau_{\mathrm{pred},O} = \sum_i T_i \big(r_i \times d_i^{\mathrm{sim}} - \chi_i c_\tau d_i^{\mathrm{sim}}\big), \tag{38}$$

where $\sigma = [1, -1, 1, -1]^T$ assigns opposite tilt offsets to groups $\{1, 3\}$ and $\{2, 4\}$ in this predictor, and $\chi = [1, 1, -1, -1]^T$. Since R_y preserves the y component, compatibility with the physical thrust plane requires

$$e_2^T d_i^{\mathrm{sim}} = e_2^T Q_\ell(\hat{q})[-\sin a_\ell, 0, -\cos a_\ell]^T = 0. \tag{39}$$

The implementation constructs this basis from the body lateral axis expressed in inertial coordinates:

$$y_\ell = \hat{R} e_2, \qquad z_* = e_3 - y_\ell (y_\ell^T e_3), \tag{40}$$

$$z_\ell = z_* / \|z_*\|, \qquad x_\ell = y_\ell \times z_\ell, \tag{41}$$

$$Q_\ell = \hat{R}^T [x_\ell \; y_\ell \; z_\ell]. \tag{42}$$

When $\|z_*\| < 10^{-6}$, the implementation substitutes $\hat{R} e_3$ for z_* before normalization. For a unit quaternion, the resulting basis is orthonormal and its second axis is e_2 in body coordinates. Consequently, $(Q_\ell)_{21} = (Q_\ell)_{23} = 0$, mapped thrust directions have unit norm, and (39) holds. Continuity through the fallback branch remains unverified; physical execution is discussed in Section VI-B.

Each call starts from $z_a^{(0)} = [T_0, T_0, T_0, T_0, 0]^T$. Trial thrusts are projected onto the bounds implied by motor PWM $[1100, 1900]$, and

$$|\beta| \le \min\{\pi/2, \max(0, 7\pi/12 - |a_\ell|)\}. \tag{43}$$

The residual-decrease test uses

$$C(z_a) = \|D_4^{-1}(y_* - \mathcal{H})\|_2^2, \qquad D_4 = \mathrm{diag}(\tau_*, \tau_*, \tau_*, F_*), \tag{44}$$

with $\tau_* = 1$ N m and $F_* = 1$ N. The solver permits ten regularized linearization iterations and eight step halvings, with a projected-gradient fallback. Steps require a decrease greater than 10^{-12}; unsuccessful searches terminate the call.

The 0.15-rad limit bounds an iteration step, not intersample tilt rate. Calls are independent; the omitted physical constraints are discussed in Section VI-B. The logged raw 4×5 iterate Jacobian is not a normalized geometric feasibility margin.

F. Actuation and Residual Definitions

Table II lists the numerical actuator settings. The allocator uses quasi-static thrust bounds of approximately 0.947–34.702 N per rotor. At the lower command boundary, the plant's motor-enable gate instead sets the target speed to zero (Appendix A-B). Thus the allocator bounds are neither exact plant-output bounds nor measured thrust-capacity limits.

Define the logged static and reconstructed dynamic residuals about O as

$$r_s = \tau_{\mathrm{req},O} - \tau_{\mathrm{pred},O}, \qquad r_d = \tau_{\mathrm{req},O} - \tau_{\mathrm{rec},O}. \tag{45}$$

The reconstruction uses simulated lever-arm, rotor-drag, and quasi-steady gyroscopic contributions. Relative to the static predictor, it also reflects motor lag, PWM quantization, and sampling differences. The S2 correction is not included as actuator output. The collective residual r_ℓ is scalar and is not a three-dimensional force residual.

The analytical moment residual uses realized-minus-commanded sign. It equals $-r_d$ only for matching definitions and reference points. For a consistently defined force residual r_F, transfer gives $r_{\tau,C} = r_{\tau,O} - c_B \times r_F$. The scalar r_ℓ cannot supply this vector correction. The terms r_s, r_d, τ_{corr}, and τ_{add} are alternative diagnostics or model inputs, not four disturbances to be summed.

IV. Conditional Local Analysis

The object of analysis is an idealized continuous-time local pitch model. It uses a single integral state and ideal derivative feedback, whereas the implemented controller has two clipped integral states, derivative-filter states, output mixing, and sampled updates. The theorem therefore applies to the model defined below; transferring it to the implemented controller requires a separate state/residual matching argument. Roll/yaw motion, observer-compensation error, and actuator mismatch are external inputs to this subsystem; their bounds are not proved here. The result is a conditional finite-time estimate, with an ultimate bound only under continued uniform conditions.

A. Reduced States and Regularity

In a consistent pure-pitch error chart, $e_\theta = \theta_d - \theta$ and $|e_\theta| \le \rho < \pi$ give

$$\eta_e^+ = \cos(e_\theta/2), \quad \omega_c(e_\theta) = \mathrm{sat}_{\bar{\omega}_c}[k_{q,y} \sin(e_\theta/2)]. \tag{46}$$

The scalar identity does not hold for a general off-axis error. For the corresponding geodesic quaternion error, $\eta_e^+ = \cos(e_R/2)$, and exclusion of the 180° attitude-error boundary $\eta_e = 0$ requires a separate condition.

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Define $e_\omega = \omega_c - \Omega_y$, $x_r = [e_\omega, e_f, \zeta]^T$, and the actual command discrepancy $\delta_c = \Omega_{c,y} - \omega_c$. The auxiliary local model is

$$\dot{e}_\theta = -\omega_c + e_\omega + d_\theta, \quad d_\theta = \dot{\theta}_d + d_g, \tag{47}$$
$$\dot{e}_\omega = -v + d_r, \quad \dot{e}_f = -\omega_f e_f + \omega_f(e_\omega + \delta_c),$$
$$\dot{\zeta} = e_f + d_I, \quad v = k_p e_f + k_i \zeta + k_d \dot{e}_f. \tag{48}$$

For pure pitch, $d_g = \delta_c = 0$. The term d_I permits a bounded continuous-time integral correction; it does not represent an unexamined discrete reset. The difference δ_c enters both the filter and derivative feedback, so it cannot be placed solely in a moment residual.

In physical-moment coordinates about O, let $\dot{\Omega}_y = b_y \tau_y + d_y$, $\tau_y = \tau_{c,y} + r_{\tau,y}$, and $\tau_{c,y} = \bar{b}_0^{-1}(v - \Gamma_y z_{2,y})$. The term d_y includes the O-to-C transfer and remaining pitch coupling, using one consistent plant equation. With $\alpha_b = b_y/\bar{b}_0$,

$$d_r = \dot{\omega}_c + (1 - \alpha_b)v - (d_y - \alpha_b \Gamma_y z_{2,y}) - b_y r_{\tau,y}. \tag{49}$$

Here $\bar{b}_0$ and the gains in v define the analytical physical-input model; they are not the normalized gains in Table I. The implemented mapping is given by (26)–(32). Its two branches, clipping, derivative filtering, and estimation errors have not been reduced to the auxiliary model with justified uniform residual bounds. The analytical scheduling coordinate $\lambda \in [0, 1]$ parameterizes k_p, k_i, k_d and is not identified with λ_a.

The argument needs θ_d to be absolutely continuous with essentially bounded rate, not bounded acceleration. The clipped ramp meets this scalar condition. Since $\omega_c(e)$ is Lipschitz, with $L_c = k_{q,y}/2$, an absolutely continuous e_θ yields an absolutely continuous command and an almost-everywhere derivative. No derivative of $\dot{\theta}_d$ or b_y is used below. The state stays continuous at the ramp corners, where the comparison inequality is integrated across a measure-zero set. This reasoning presumes Carathéodory solutions, an absolutely continuous schedule, and no quaternion sign switch in the chosen chart. It does not cover unmodeled state jumps. This scalar regularity argument does not establish the required regularity of the complete online reference or the digital implementation.

B. Candidate-Domain Bounds

Choose $\mathcal{K} = \{|e_\theta| \le \rho, \ \|x_r\| \le r\}$. Assume bounds on d_g, δ_c, d_I, the observer-compensation term, and moment mismatch that hold uniformly on $\mathcal{K}$ and a separately justified external-input set. Write

$$|d_\theta| \le \bar{d}_\theta, \qquad |\delta_c| \le \bar{\delta}_c, \qquad |d_I| \le \bar{d}_I,$$
$$0 < \underline{b} \le b_y \le \bar{b}, \qquad |1 - \alpha_b| \le \delta_b,$$
$$|d_y - \alpha_b \Gamma_y z_{2,y}| \le \Delta_o, \qquad |r_{\tau,y}| \le \bar{r}_\tau. \tag{50}$$

An external set containing other closed-loop states requires its own invariance argument; it cannot be justified by this pitch result. These are domain bounds, not maxima taken from a finite record.

Let $a_u = [k_d\omega_f, k_p - k_d\omega_f, k_i]$, $L_u = \max_\lambda \|a_u\|$, and $\bar{k}_d = \max_\lambda |k_d|$. Candidate-domain bounds are

$$\bar{v}_\mathcal{K} = L_u r + \bar{k}_d \omega_f \bar{\delta}_c,$$
$$W_\mathcal{K} = \min\{\bar{\omega}_c, k_{q,y} \sin(\rho/2)\},$$
$$\bar{d}_{r,\mathcal{K}} = L_c(W_\mathcal{K} + r + \bar{d}_\theta) + \delta_b \bar{v}_\mathcal{K} + \Delta_o + \bar{b}\bar{r}_\tau. \tag{51}$$

Thus input and integral-state dependence in d_r is retained through r, before proving that the solution stays in $\mathcal{K}$.

The rate subsystem is

$$\dot{x}_r = A_r(\lambda)x_r + w_r, \quad w_r = B_r d_r + E_r \delta_c + B_I d_I,$$
$$A_r = \begin{bmatrix} -k_d\omega_f & k_d\omega_f - k_p & -k_i \\ \omega_f & -\omega_f & 0 \\ 0 & 1 & 0 \end{bmatrix}, \tag{52}$$
$$B_r = [1, 0, 0]^T, \quad B_I = [0, 0, 1]^T,$$
$$E_r = [-k_d\omega_f, \omega_f, 0]^T, \quad \|w_r\| \le D_\mathcal{K},$$
$$D_\mathcal{K} = \bar{d}_{r,\mathcal{K}} + \max_\lambda \|E_r\| \bar{\delta}_c + \bar{d}_I. \tag{53}$$

Norms refer to consistently scaled analytical states. The fixed-λ polynomial is $s^3 + \omega_f(1 + k_d)s^2 + \omega_f k_p s + \omega_f k_i$. Positive gains satisfy its Routh–Hurwitz test when $\omega_f k_p(1 + k_d) > k_i$; this alone does not address scheduling.

Assume a differentiable Lyapunov family with

$$A_r^T P_r + P_r A_r = -Q_r, \quad \underline{p} I \preceq P_r \preceq \bar{p} I,$$
$$Q_r \succeq \underline{q} I, \quad \|\partial_\lambda P_r\| \le L_P,$$
$$|\dot{\lambda}| \le \bar{\lambda}_1, \quad \alpha_r = \underline{q} - L_P \bar{\lambda}_1 > 0. \tag{54}$$

The bound on $\dot{\lambda}$ is an assumption on the analytical gain family. The implemented weight λ_a depends on both estimated attitude and the common-tilt command. Its sampled and switched behavior cannot be substituted into this continuous-time condition without an additional matching argument. For $V_r = x_r^T P_r x_r$, differentiation and Young's inequality give

$$\dot{V}_r \le -\tfrac{1}{2}\alpha_r \|x_r\|^2 + 2\bar{p}^2 D_\mathcal{K}^2/\alpha_r \tag{55}$$

almost everywhere before exit from $\mathcal{K}$.

C. Finite-Time Estimate and Conditional Continuation

Let $k_o = \min_{0<|e|\le\rho} \omega_c(e)/e > 0$, with the continuous extension at zero. Choose $c_r > 2/(k_o \alpha_r)$ and define

$$V_c = e_\theta^2/2 + c_r V_r, \quad x_c = [e_\theta, x_r^T]^T,$$
$$\alpha_c = \min\{k_o/2, c_r\alpha_r/2 - 1/k_o\},$$
$$\underline{c}_V = \min\{1/2, c_r\underline{p}\}, \quad \bar{c}_V = \max\{1/2, c_r\bar{p}\},$$
$$a = \alpha_c/\bar{c}_V, \quad \Delta = \bar{d}_\theta^2/k_o + 2c_r\bar{p}^2 D_\mathcal{K}^2/\alpha_r,$$
$$\mathcal{B}(t; t_0, V_0, \Delta) = e^{-a(t-t_0)}V_0 + \frac{\Delta}{a}(1 - e^{-a(t-t_0)}). \tag{56}$$

Theorem 1 (Conditional local finite-time pitch bound)**.** *Suppose the stated reduction, regularity, external-input bounds, and Lyapunov family hold on a finite interval $[t_0, T]$. If some V_T satisfies*

$$\max\{V_c(t_0), \Delta/a\} < V_T < \min\{\rho^2/2, c_r\underline{p}r^2\}, \tag{57}$$

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then the reduced state remains in $\mathcal{K}$ throughout that interval and

$$V_c(t) \le \mathcal{B}(t; t_0, V_c(t_0), \Delta),$$
$$|e_\theta(t)| \le \sqrt{2\mathcal{B}(t)}, \quad \|x_r(t)\| \le \sqrt{\mathcal{B}(t)/(c_r \underline{p})}. \tag{58}$$

Proof. The sector property yields $e_\theta \dot{e}_\theta \le -k_o e_\theta^2/2 + \|x_r\|^2/k_o + \bar{d}_\theta^2/k_o$. Combining with (55) gives $\dot{V}_c \le -\alpha_c \|x_c\|^2 + \Delta \le -aV_c + \Delta$ almost everywhere up to a possible first exit. Comparison gives (56). By (57), this value is strictly below V_T, whose sublevel lies inside $\mathcal{K}$ for every λ. Continuity rules out the first exit. The quadratic lower bounds yield (58). □

At reference completion, (58) gives $|e_\theta(t_f)| \le \sqrt{2\mathcal{B}(t_f)}$. On a subsequent hold interval, the argument restarts at t_f, using $V_c(t_f) \le \mathcal{B}(t_f)$ and a justified hold-stage bound Δ_h. Setting the scalar reference rate to zero does not remove other forcing. If all conditions continue uniformly for unbounded time, $\limsup \|x_c\| \le \sqrt{\Delta/(a\underline{c}_V)}$ follows. Neither continuation nor numerical satisfaction of (57) has been established for the recorded runs.

Even under exact pure pitch, correspondence with (61) requires a common unwrapped phase chart. In that case,

$$N(t_f) - 1 = \frac{e_\theta(t_r) - e_\theta(t_f)}{2\pi}, \tag{59}$$

so the initial phase error also enters the endpoint test. With off-axis references, a bound on the additional projection discrepancy is required. No numerical $N(t_f)$ or full-attitude bound is inferred from this theorem. Sampling, filtering, clipping, and quantization enter this continuous-time model only through justified matching equations or uniform residual bounds; digital state jumps require a separate analysis.

V. Numerical Evaluation

A. Settings and Paired Comparisons

Two S1/S2 simulation pairs disable or enable the prescribed coupling module in (10)–(18). Each pair uses common controller and allocator settings, with no retuning or added randomized disturbance.

The *strict common-reference pair* replays the online-S1 complete desired-quaternion history in both cases. Attitude feedback, x-position feedback through the common-tilt servo command, and real-time altitude feedback generating P_0 remain active. The replayed quaternion replaces the online roll reference generated by the y-position channel. This pair measures the prescribed coupling effect within that replay configuration.

The *online-reference pair* retains the feedback reference generator in both cases. The scalar pitch task and pilot commands are common, while $q_{d,\mathrm{on}}$ can differ with the feedback state. Its S2-minus-S1 changes therefore include the response of reference generation to the imposed coupling. Coupling increments are calculated within each pair. Differences between mechanisms are discussed as differences in closed-loop configuration, rather than formed by combining an S1 case from one pair with an S2 case from the other.

TABLE II
Simulation and Actuator-Model Settings

Item	Value
Record; evaluation window	0–65 s; 20–65 s
Plant/controller/INS period	0.002 s
Position-command period	0.004 s
Solver / fixed step	`ode1` / 0.002 s
Sampled pitch interval	20–51.416 s
Mass-property transition	27–45 s
Mass m	2.288 kg
Inertia J_0 (kg m^2)	$\mathrm{diag}(0.082, 0.082, 0.149)$
Final c_B; inertia multiplier	$(3, 0, 2)$ cm; 1.10
C_t; C_m	1.64×10^{-5}; 3.41×10^{-7}
k_ω; ω_b	1518; 88.448 rad/s
Motor time constant	0.02 s
Motor–propeller inertia	1.287×10^{-4} kg m^2
Motor PWM bounds	1100–1900
Tilt PWM encoding bounds	500–2500
Solver iterations / halvings	10 / 8
Per-iteration β step	0.15 rad
Branch / encoding-angle limits	$\pi/2$ / $7\pi/12$ rad

The records span 0–65 s, including startup; RMS and peaks use $\mathcal{T}_e = [20, 65]$ s. The 13.584 s after t_f exceed the specified minimum 10-s settling record. Both S2 configurations use the profiles in (10)–(12).

B. Metrics and Performance Comparisons

The scalar-first quaternions are normalized and continued in sign. The attitude and phase metrics use the simulated INS quaternion $\hat{q}$ supplied to feedback, rather than the plant truth quaternion. The geodesic attitude error is

$$e_R = 2 \arccos\big(\mathrm{clip}(|q_d^T \hat{q}|, 0, 1)\big). \tag{60}$$

For the recorded attitude convention, the unwrapped y-twist and accumulated pitch revolutions are

$$\widetilde{\theta} = 2\,\mathrm{unwrap}\big(\mathrm{atan2}(\hat{q}_y, \hat{q}_w)\big), \qquad N(t) = \frac{\widetilde{\theta}(t) - \widetilde{\theta}(20)}{2\pi}. \tag{61}$$

This is a task-specific projection, not a general integer-valued topological invariant of three-dimensional attitude paths. The present definition uses the logged $\hat{q}_w, \hat{q}_y$ directly and is not extended here to arbitrary initial attitude conventions. The projection norm $\sqrt{\hat{q}_w^2 + \hat{q}_y^2}$ and off-axis swing are reported to assess its suitability for these records.

Position is evaluated at O. The evaluation target is $x_d = y_d = 0$ and $h_d = h(20)$, giving

$$e_p = [x, y, h - h(20)]^T. \tag{62}$$

This plant-state evaluation target differs from the latched INS targets in (19); the resulting offsets are retained in the reported errors. For a signal z on $[t_1, t_2]$, RMS is

$$z_{\mathrm{RMS}} = \left(\frac{1}{t_2 - t_1} \int_{t_1}^{t_2} \|z(t)\|^2 dt \right)^{1/2}, \tag{63}$$

computed by trapezoidal integration. Peaks use every standardized sample in the specified window.

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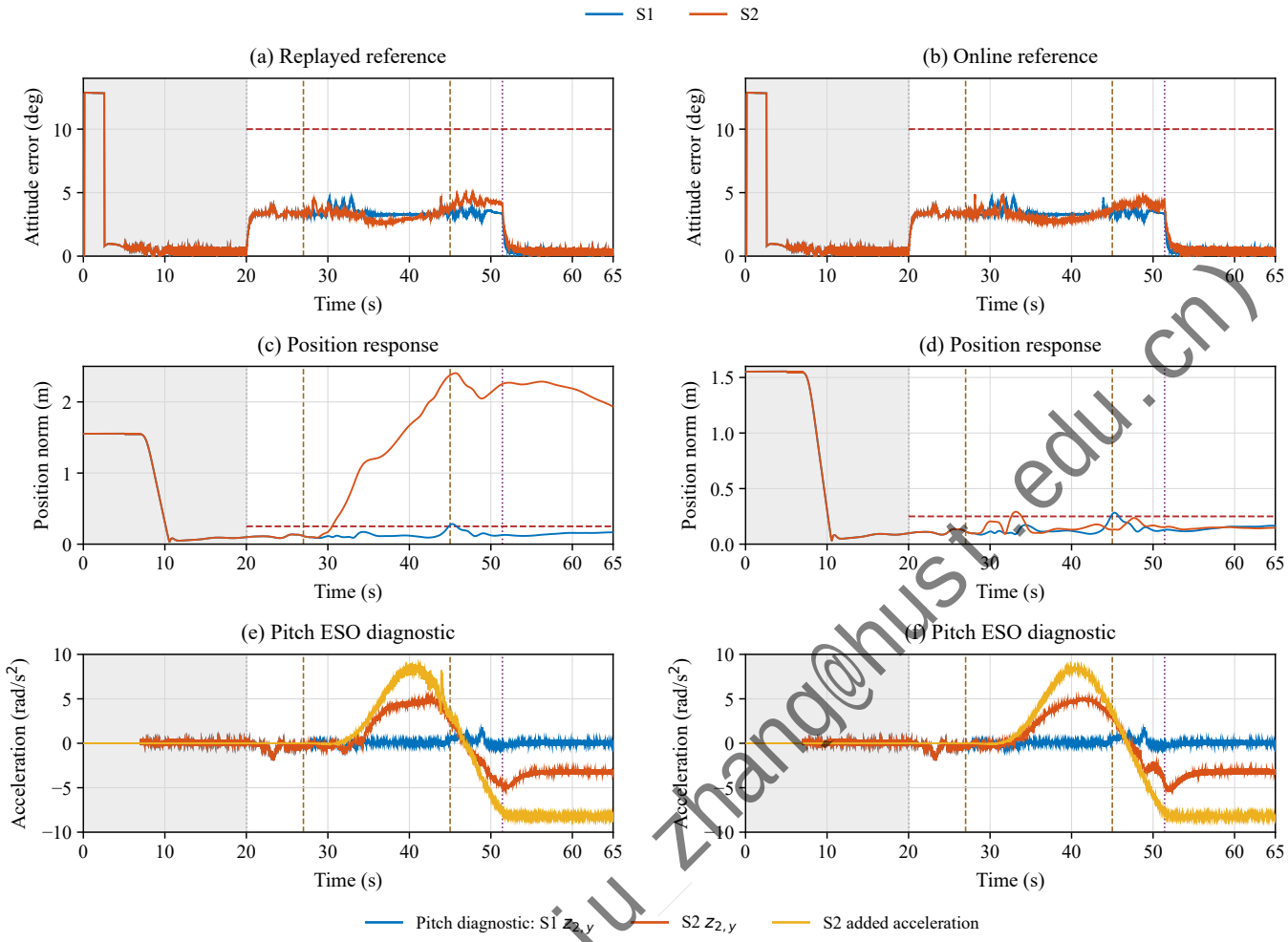


Fig. 2. Simulation responses under complete-reference replay (left) and online reference generation (right). Rows show geodesic attitude-estimate error, position-error norm, and the pitch ESO diagnostic. The gray 0–20 s startup region is outside the evaluation window; dashed limits apply on 20–65 s. Position axes differ: 0–2.5 m on the left, 0–1.6 m on the right. The last row compares total disturbance estimate $z_{2,y}$ with the S2 added acceleration $(J_0^{-1}\tau_{\mathrm{add}})_y$; the coupling acceleration is distinct from the actuator-only dynamic residual. These are simulation diagnostics, not measured actuation or observer-error traces.

The specified numerical conditions require finite critical signals, normal termination, at least 10 s after reference completion, and

$$e_{R,\mathrm{pk}} \le 10^\circ, \qquad \|e_p\|_{\mathrm{pk}} \le 0.25\ \mathrm{m}, \qquad |N(t_f) - 1| \le 0.01. \tag{64}$$

The first two peaks use 20–65 s; the revolution-error test is at $t_f = 51.416$ s. These performance conditions are separate from physical actuator feasibility and from mathematical UUB.

C. Attitude and Position Responses

Table III and Fig. 2 show small changes in attitude-estimate tracking under the prescribed perturbation. Online S2 increases attitude-estimate RMS/peak by $0.0252^\circ/0.2643^\circ$ and position peak by 0.0080 m within its pair. This modest position-peak increment indicates limited additional position error in the tested online configuration. The absolute peaks, 0.2913 m for S2 and 0.2834 m for S1, remain above the specified 0.25-m condition. Under replay, attitude RMS/peak increase by $0.0785^\circ/0.2105^\circ$, while position RMS reaches 1.693 m and peak reaches 2.406 m, a 2.1222-m peak increase. The largest contribution is lateral: S2 y-error RMS is 1.6869 m and its reported peak is approximately 2.384 m.

These are within-pair responses to the prescribed module. Similar attitude-error changes coexist with different position responses, but the replay drift is specific to a configuration bypassing online roll-reference adjustment. Cross-mechanism differences do not isolate controller superiority or a general consequence of manipulator motion.

D. Revolution Progress and Reference-End Error

Figure 3 retains the full unwrapped phase and simulated body-y rate histories. Starting from the 0° anchor at 20 s, all four records first reach 90°, 180°, 270°, and 360° in that order. The metric-processing code assigns the zero-phase anchor directly; it detects each positive threshold using a 10^{-8}-degree tolerance and linearly interpolates the neighboring samples. Three-decimal crossing times describe that interpolant on the 2-ms grid, not millisecond physical timing accuracy. The min-

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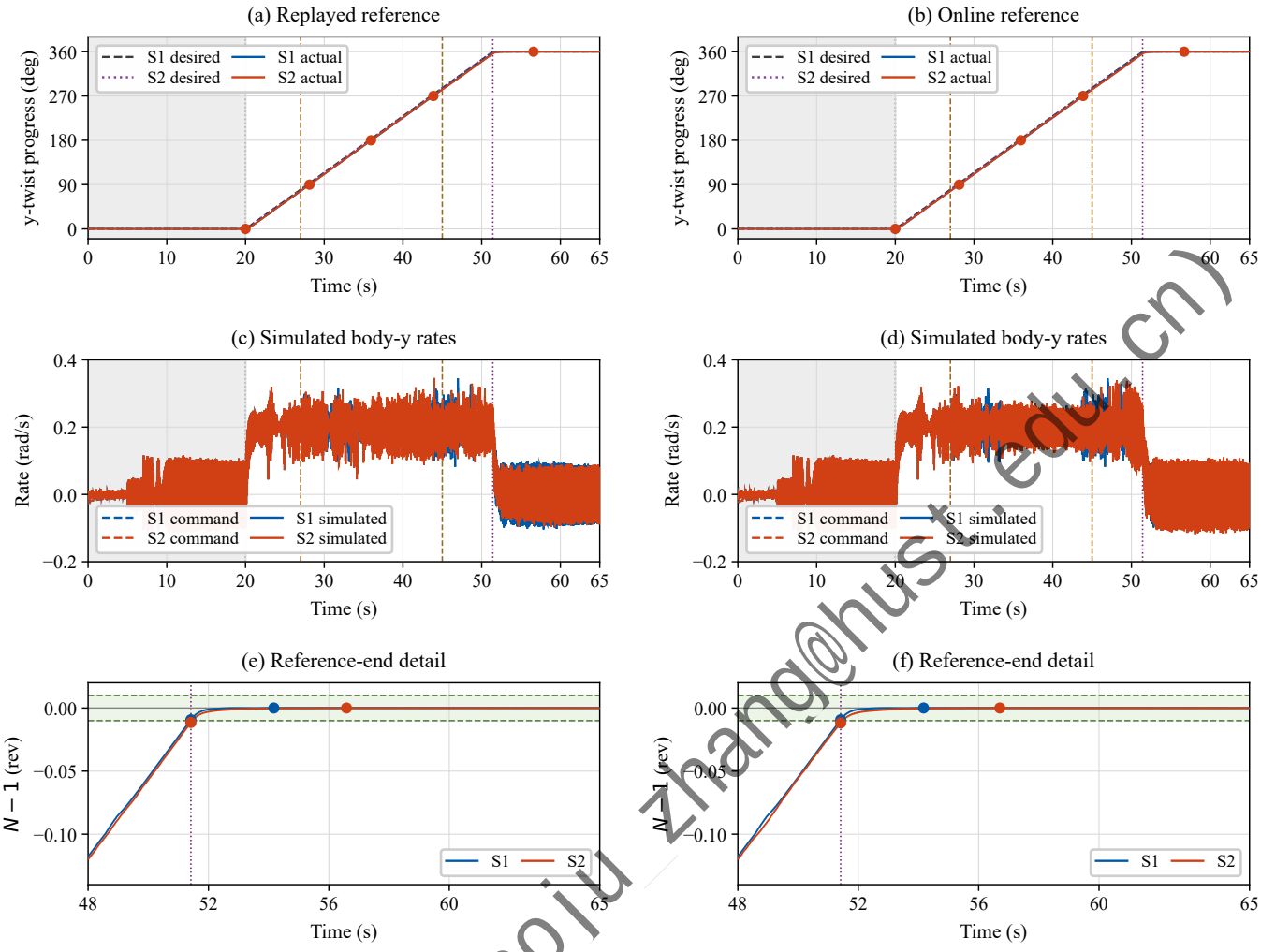


Fig. 3. Path and timing evidence for replay (left) and online reference generation (right). The first two rows retain 0–65 s: accumulated y-twist phase from the INS attitude estimate, inner-loop rate commands, and simulated INS body-y rates. Phase dots denote S2's assigned zero anchor and four interpolated threshold crossings. The last row enlarges 48–65 s, with the ± 0.01-rev band and the check at $t_f = 51.416$ s; later zero crossings are separate.

TABLE III
PAIRED TRACKING RESULTS: RMS AND PEAKS ON 20–65 S

Metric (unit)	Strict common reference		Online reference		Specified condition
	S1	S2	S1	S2	
Attitude-estimate error RMS (deg)	2.844	2.923	2.844	2.869	–
Attitude-estimate error peak (deg)	4.622	4.833	4.622	4.886	≤ 10: all pass
Position error RMS, y (m)	0.07486	1.68690	0.07486	0.08686	–
Position norm RMS (m)	0.13584	1.69322	0.13584	0.15139	–
Position norm peak (m)	0.2834	2.4056	0.2834	0.2913	≤ 0.25: all fail
$N(t_f)$ (rev)	0.990669	0.988571	0.990669	0.988166	–
$\lvert N(t_f) - 1 \rvert$ (rev)	0.009331	0.011429	0.009331	0.011834	≤ 0.01: S1 pass; S2 fail
First 360° crossing (s)	54.171	56.586	54.171	56.704	Later traversal
$N(65)$ (rev)	1.000006	0.999990	1.000006	0.999984	Settling value
Maximum off-axis swing (deg)	3.110	3.102	3.110	3.287	–
No case meets all specified conditions; individual comparisons are retained above.					

imum y-twist projection norm exceeds 0.9995 and maximum off-axis swing is below 3.29° on 20–65 s, supporting this projection for these records.

At t_f, the S1 revolution error is 0.009331 rev and satisfies the 0.01-rev condition in both pairs. Strict-reference and online-reference S2 errors are 0.011429 rev and 0.011834 rev; both exceed the specified 0.01-rev condition. Nevertheless, their recorded attitude estimates complete the traversal: their

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TABLE IV
ALLOCATION RESIDUALS ON 20–65 S

Metric	S1 in each pair	Strict-reference S2	Online-reference S2
Static RMS (x, y, z) (N m)	$(0.0561, 0.0252, 0.00628)$	$(0.0462, 0.0186, 0.00600)$	$(0.0521, 0.0227, 0.00634)$
Static peak (x, y, z) (N m)	$(0.8457, 0.6591, 0.1628)$	$(0.7879, 0.4706, 0.1659)$	$(0.9651, 0.6570, 0.1619)$
Dynamic RMS (x, y, z) (N m)	$(0.6014, 0.4978, 0.0460)$	$(0.4934, 0.4895, 0.0514)$	$(0.5775, 0.5173, 0.0467)$
Dynamic peak (x, y, z) (N m)	$(1.6322, 1.9510, 0.4584)$	$(1.5423, 1.8669, 0.4377)$	$(1.8922, 2.0510, 0.3782)$
Collective RMS (N)	0.01007	0.01033	0.01156
Collective peak (N)	0.60449	0.60449	0.60449

first interpolated 360° crossings occur later, at 56.586 s and 56.704 s; S1 crosses at 54.171 s. The 65-s values characterize subsequent settling and do not replace the reference-end checks.

The phase has local reversals. Accumulated reverse progress over 20–65 s is 19.558° for S1, 12.640° for strict-reference S2, and 15.538° for online-reference S2. Ordered crossings therefore establish traversal without implying strict monotonicity.

As a separate cross-check, define

$$N_\Omega(t) = \frac{1}{2\pi} \int_{20}^{t} \Omega_{y,\mathrm{sim}}(\xi)\, d\xi. \tag{65}$$

The differences $N(t_f) - N_\Omega(t_f)$ are 0.001048 rev for S1, 0.000442 rev for strict-reference S2, and 0.001206 rev for online-reference S2. These measures need not coincide during three-axis motion. The body-rate signals are simulation outputs. The inner-loop rate command also differs from the derivative of the scalar pitch ramp.

E. Allocation and Observer Diagnostics

Table IV separates the static solver residual from the reconstructed dynamic residual. The latter is larger in both pairs, so the optimizer residual alone does not represent the moment realized by the actuator model. No validated residual threshold is available; these entries are reported as magnitudes rather than pass/fail tests.

The simulation plant receives motor commands and replacement simulation-tilt commands; the original hardware-tilt channels do not supply its active tilt inputs. Channel numbering and boundary-occupancy statistics are given in Appendix A-B. Command-boundary occupancy does not establish physical saturation or loaded servo performance.

The final row of Fig. 2 displays $z_{2,y}$, an estimate of total equivalent disturbance, alongside $(J_0^{-1}\tau_{\mathrm{add}})_y$, the S2 module's incremental plant acceleration. The two signals have different roles. Their proximity over part of the record does not provide an observer-error bound or isolate an estimate of arm coupling. All of these results use fixed-gain ESOs.

VI. DISCUSSION AND LIMITATIONS

A. Interpreting the Paired Results

The paired results provide encouraging preliminary evidence of attitude-estimate tracking under the prescribed mass-property variation: both S2 peak errors remain below 5°, with only modest changes from S1. The online configuration also retains a position-error peak close to its nominal value. These observations concern the tested conditions and the logged attitude estimates, rather than a general robustness guarantee. Position regulation additionally depends on the resulting force direction and on the evaluation target. The S2-minus-S1 peak increases of 2.1222 m under replay and 0.0080 m online describe additional error in their respective configurations, not all of the error in S2. Replay retains the x-position feedback through common tilting but removes the current y-position error from roll-reference generation; the online configuration retains that reference feedback. This architectural difference is a plausible contributor to the different position responses. The componentwise metrics identify lateral error as the largest contribution in the replay case, but the comparisons do not isolate the quantitative contribution of each feedback path. The prescribed center-of-mass and inertia changes occur together, so their individual effects are not separated.

The outer loop has no reference-rate feedforward. Under exact pure pitch, zero attitude error gives $\omega_c = 0$, whereas the constant-rate task requires nonzero rate. With an unsaturated outer loop and ideal rate tracking, a constant-lag balance would be $k_{q,y} \sin(e_\theta^*/2) = \omega_r$. With $k_{q,y} = 7$ and $\omega_r = 0.2$ rad/s, this ideal balance gives $e_\theta^* = 2 \arcsin(0.2/7) \approx 3.27^\circ$. Transients, off-axis motion, and actuation also contribute; the formula is a conditional mechanism explanation, not a fit to the endpoint data.

B. Model and Implementation Scope

The constant-mass model prescribes center-of-mass and inertia histories with zero relative internal angular momentum. Identified joint dynamics, contact, and payload variation are absent. The unit-quaternion basis construction satisfies the planar-thrust condition, but physical execution additionally requires a validated servo mapping, identified thrust capacity, finite tilt travel and rate, clearance, and acceptable electrical/thermal loads. The allocator omits intersample branch-continuity, tilt-rate, collision, and power constraints. These requirements have not been jointly verified, and no full-revolution flight has been performed. A future allocator could retain actuator state and impose branch continuity, tilt-rate, and collision constraints; it would require separate implementation and evaluation.

C. Analytical Scope and Reproducibility

The finite-time estimate in Section IV applies to an auxiliary continuous-time model; a matching argument for the two-branch digital controller and numerical verification of the

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domain assumptions remain absent. It therefore does not certify full-attitude, position, or discrete-time stability.

Section III and Appendix A specify the feedback laws, discrete updates, and actuator-command mappings. Simulation INS estimates drive the controller, while position evaluation uses plant truth and a different target; these distinctions are essential when interpreting the errors.

Future work will separate center-of-mass and inertia variations to quantify their individual contributions under each reference mechanism. Comparisons at additional pitch rates will also help assess the predicted rate-dependent tracking lag.

VII. Conclusion

This preliminary simulation study examined a full-pitch task under prescribed equivalent mass-property variation. Peak geodesic errors of the simulated attitude estimates remained below 5° in both perturbed configurations: 4.833° under replay and 4.886° with online reference generation. The online position-error peak increased by 0.0080 m from its nominal value to 0.2913 m, indicating limited additional position error for this perturbation. Both perturbed attitude-estimate records subsequently traversed the complete revolution. Together, these observations provide encouraging numerical evidence for the proposed architecture in the tested scenario.

The contrasting 2.406-m position peak under replay highlights the sensitivity of position regulation to the reference-feedback configuration. Completion delay and position excursions remain limitations, with the specified threshold comparisons reported in Section V. The auxiliary continuous-time pitch model supplies a conditional error bound, not a stability certificate for the implemented digital controller. The study establishes an initial numerical basis for further investigation; plant-truth attitude validation, broader operating conditions, and physical actuator feasibility remain to be evaluated.

Appendix A
Discrete Realization and Command Interfaces

A. Velocity-Command and PID Filters

For each component of a velocity command v_c, the tracking filter maintains x_1, x_2, initialized to zero, and uses forward-Euler updates

$$x_1^+ = x_1 + hx_2, \qquad x_2^+ = x_2 + hf_{\mathrm{han}}(x_1 - v_c, x_2; r_0, h_0). \tag{66}$$

Its output is x_1. Here $h_0 = 0.05$ s, with $r_0 = 6(9.806)$ for each horizontal component and $r_0 = 8(9.806)$ vertically. The implemented nonlinear function is defined by

$$\begin{aligned} d &= r_0 h_0^2, \quad a_0 = h_0 x_2, \quad y = e + a_0, \\ a_1 &= \sqrt{d(d + 8|y|)}, \quad a_2 = a_0 + \tfrac{1}{2}\operatorname{sgn}(y)(a_1 - d), \\ s_y &= \tfrac{1}{2}[\operatorname{sgn}(y + d) - \operatorname{sgn}(y - d)], \\ A &= a_2 + (a_0 + y - a_2)s_y, \\ s_A &= \tfrac{1}{2}[\operatorname{sgn}(A + d) - \operatorname{sgn}(A - d)], \\ f_{\mathrm{han}} &= -r_0(A/d - \operatorname{sgn} A)s_A - r_0 \operatorname{sgn} A. \end{aligned} \tag{67}$$

This filter runs at the controller period $h = 0.002$ s, after the 4-ms position-command update.

For velocity PID input e, the discrete realization is $u = k_p e + I + D$, with $I^+ = \mathrm{clip}(I + hk_i e, I_{\min}, I_{\max})$, $\xi^+ = \xi + h\omega_d(e - \xi)$, and $D = \mathrm{clip}(k_d\omega_d(e - \xi), D_{\min}, D_{\max})$. Here $\omega_d = 2\pi(10)$ rad/s. Reset initializes the derivative-filter state to its input and forces its output to zero. Integral reset values and limits are specified in Section III; the vertical derivative limits are ± 0.1, although its derivative gain is zero. The 30-Hz rate-error filter is initialized to its input; the rate integrators and derivative filters use the reset rules in (25). The integrator reset is level-sensitive; the flight-management reset flag is zero during the evaluated armed position-hold interval. There is no separate error-dependent gain adaptation in the active PID or ESO blocks.

B. PWM Encoding and Actuator Dynamics

PWM1–4 encode motor speed. PWM5–8 are hardware-tilt command channels retained as diagnostics; PWM9–12 supply the active simulation-tilt inputs. Let $k_s = 12000/(7\pi)$ and define

$$w_a = [1, -1, -1, 1]^T, \qquad w_b = [1, 1, -1, -1]^T. \tag{68}$$

The active tilt encoding is

$$\begin{aligned} P^*_{\mathrm{tilt}} &= 1500\mathbf{1} + k_s(w_a a_\ell + w_b \beta), \\ P_{9:12} &= \mathrm{clip}(\lfloor P^*_{\mathrm{tilt}} \rfloor, 500, 2500)\,. \end{aligned} \tag{69}$$

The hardware-diagnostic encoding uses $a_\ell - \hat{\theta}$ in place of a_ℓ, where $\hat{\theta}$ is the INS Euler pitch. The plant decodes the active channels as

$$a_c = \frac{w_a^T P_{9:12}}{4k_s}, \qquad \beta_c = \frac{w_b^T P_{9:12}}{4k_s}. \tag{70}$$

Both sign vectors have zero sum, so the 1500 offset cancels. Without quantization and clipping, (70) exactly inverts (69). These are virtual common/differential coordinates; they do not by themselves calibrate physical servo angles.

For each $v \in \{a, \beta\}$, the actuator model rate-limits the tracking discrepancy, not the angle itself:

$$\begin{aligned} r_v[k] &= v_c[k] - v[k], \\ \ell_v[k] &= \ell_v[k-1] + \mathrm{sat}_{hL}(r_v[k] - \ell_v[k-1])\,, \\ v[k+1] &= v[k] + (h/0.02)\ell_v[k], \qquad L = 1714.29. \end{aligned} \tag{71}$$

The states start at zero. Thus the stored limiter setting is not a validated loaded servo-speed bound. The controller-to-plant command connection includes one 2-ms delay. The actuator wrench calculation also uses a one-step-delayed plant quaternion, initialized to $[1, 0, 0, 0]^T$.

Motor PWM is clipped and converted to unsigned integer values with floor rounding. For the plant input P_i, the motor model uses

$$\begin{aligned} \omega_{i,c} &= \begin{cases} 1518(P_i - 1000)10^{-3} + 88.448, & P_i > 1100, \\ 0, & P_i \le 1100, \end{cases} \\ \omega_i[k+1] &= \omega_i[k] + (h/0.02)(\omega_{i,c}[k] - \omega_i[k]). \end{aligned} \tag{72}$$

The allocator's lower quasi-static thrust bound is approximately 0.947 N, but the plant's strict motor-enable gate switches the target speed to zero at exactly 1100. This

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TABLE V
S2 COMMAND-BOUNDARY OCCUPANCY ON 20–65 S (%)

Channels / statistic	Replay	Online
PWM1–4 / range	1.262–1.760	0.938–1.440
PWM5–8 / maximum	0.324	0.00889

command-boundary discrepancy is part of the simulated actuator model.

With the lagged a, β and delayed plant attitude, the rotor model evaluates (4) using the directions in (38) and adds the quasi-steady gyroscopic term

$$H = -J_m \sum_i \chi_i \omega_i d_i, \qquad \tau_{a,O} = \tau_{\mathrm{static},O} - \Omega \times H. \quad (73)$$

Here $J_m = 1.287 \times 10^{-4}\ \mathrm{kg\,m^2}$. The coupling correction is added separately as in (18). The allocator receives $Q_\ell(\hat{q})$, whereas the plant uses its delayed true attitude in the corresponding basis construction; these signals are not interchangeable when estimating dynamic allocation mismatch.

C. Command-Boundary Statistics

Table V reports S2 boundary occupancy on 20–65 s. Motor entries give the range across channels and hardware-tilt entries the largest channel occupancy. These command statistics do not establish physical actuator saturation, travel, or loaded speed feasibility.

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